

CLINICAL AND POPULATION SCIENCES

Carotid Stenting Versus Endarterectomy for Asymptomatic Carotid Artery Stenosis: A Systematic Review and Meta-Analysis

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BACKGROUND: To compare safety and efficacy between carotid artery stenting (CAS) and carotid endarterectomy (CEA) in treating asymptomatic carotid artery stenosis based on results from randomized controlled trials.

METHODS: Randomized controlled trials comparing CAS and CEA in treating asymptomatic carotid artery stenosis were searched from databases of the EMBASE, PubMed, MEDLINE, and Cochrane libraries. Two independent reviewers identified eligible studies, extracted relevant data, and used the Cochrane risk of bias tool to assess quality. Mantel-Haenszel method random-effects models were used to estimate odds ratio (OR) regarding perioperative risks between CAS and CEA. Kaplan-Meier curve data were extracted and analyzed through Exp[(O-E)/Var] fixed-effect models to calculate the Peto odds ratio (OR) regarding long-term outcomes.

RESULTS: Sixteen articles from 7 randomized controlled trials were included, reporting relevant outcomes for 7118 asymptomatic carotid artery stenosis patients (CAS: n=3920; CEA: n=3198). Compared with the CEA group, CAS group had no difference in perioperative composite end point events including stroke, death, and myocardial infarction (MI; OR, 1.13 [95% CI, 0.87–1.47]; $P=0.37$, $I^2=0\%$). Compared with CEA, CAS had a higher risk of any stroke during the perioperative period (OR, 1.62 [95% CI, 1.16–2.24]; $P=0.004$, $I^2=0\%$) and an increased risk of nondisabling stroke (OR, 1.81 [95% CI, 1.23–2.65]; $P=0.003$, $I^2=0\%$), but there was no significant difference in disabling stroke and death between groups (OR, 0.91 [95% CI, 0.50–1.65]; $P=0.76$, $I^2=0\%$). For long-term outcomes, no difference regarding the composite outcome of any stroke, death, and myocardial infarction existed between CEA and CAS (Peto OR, 1.18 [95% CI, 0.94–1.48]; $P=0.14$, $I^2=0\%$). Individual-level patient data would be important to verify the long-term outcome results.

CONCLUSIONS: When treating asymptomatic carotid artery stenosis, CAS has comparable perioperative and long-term composite outcomes compared with CEA. However, CAS may have a higher risk of any stroke and nondisabling stroke in the perioperative period.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: carotid artery ■ carotid stenosis ■ endarterectomy, carotid ■ myocardial infarction ■ stent

Each year, there are ≈6094000 strokes worldwide.¹ Overall, about 10% to 15% of all ischemic strokes are due to a previously asymptomatic >50% carotid stenosis.² The management of carotid stenosis can involve revascularization, either with

carotid endarterectomy (CEA) or carotid artery stenting (CAS), as well as best medical therapy (BMT). Whether CEA or CAS is a better revascularization choice for asymptomatic carotid stenosis patients remains controversial. The 2021 European Stroke

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Nonstandard Abbreviations and Acronyms

ACST-2	Second Asymptomatic Carotid Surgery Trial
ACT-1	Asymptomatic Carotid Trial
BMT	best medical therapy
CAS	carotid artery stenting
CEA	carotid endarterectomy
CREST	The Carotid Revascularization Endarterectomy Versus Stenting Trial
SAPPHIRE	Stenting and Angioplasty With Protection in Patients at High Risk for Endarterectomy
SPACE-2	Stent-Protected Angioplasty in Asymptomatic Carotid Artery Stenosis vs Endarterectomy

Organization³ guidelines recommended CEA in asymptomatic carotid artery stenosis patients with stenosis $\geq 60\%$ and at increased risk of stroke with BMT alone but recommended against CAS as a routine alternative to BMT. A Cochrane systematic review⁴ showed that compared with CEA, CAS had no significant increase in perioperative death or stroke for asymptomatic carotid artery stenosis (odds ratio [OR], 1.72 [95% CI, 1.00–2.97]; $P=0.05$, $I^2=0\%$). Also, the risk of perioperative death or ipsilateral stroke was comparable between the 2 groups (OR, 1.27 [95% CI, 0.87–1.84]; $P=0.22$, $I^2=0\%$).⁴ However, analysis of long-term efficacy was unavailable due to limited evidence at that time. Recently, ACST-2 (Second Asymptomatic Carotid Surgery Trial), the largest randomized controlled trial (RCT) in asymptomatic patients, reported the 5-year follow-up outcomes, showing similar risk of nonprocedural stroke between CEA and CAS (5.3% versus 4.5, risk ratio [RR], 1.16 [95% CI, 0.86–1.57]; $P=0.33$). ACST-2 had approximately the same number of cases as all previous studies combined.⁵

Therefore, this review added new RCT evidence to compare the short-term and long-term outcomes of CAS and CEA in patients with asymptomatic carotid stenosis.

METHODS

Search Strategy

The data that support the findings of this study are available from the corresponding author upon reasonable request. We performed systematic review and meta-analysis based on the PRISMA guidelines (Preferred Reporting Items for Systematic Reviews and Meta-Analyses).⁶ The PRISMA checklist is detailed in the [Supplemental Material](#). The review was registered in the International Prospective Register of Ongoing Systematic Reviews (URL: <https://www.crd.york.ac.uk/PROSPERO/>;

Unique identifier: CRD42021276304). Ethical approval and informed consent are not applicable for this article.

We systematically searched EMBASE (via Ovid), PubMed, and Cochrane Library, from January 2000 to November 2021 for RCTs comparing CAS and CEA in asymptomatic carotid artery stenosis patients. The following medical subject terms were utilized for the online search: “randomized trial,” “asymptomatic carotid stenosis,” “carotid endarterectomy,” and “carotid angioplasty±stenting.” Our search algorithm is detailed in [Table S1](#). We only included research published in English.

Study Selection

Only RCTs evaluating participants undergoing CAS versus CEA in asymptomatic carotid artery stenosis patients were included. Outcome events included any stroke (whether disabling, severe, nondisabling, or minor), death, or myocardial infarction (MI). The exclusion criteria included observational studies, systematic reviews and meta-analyses, letters to editors, reviews, editorials, animal model studies, and basic scientific research. In addition, abstracts from conference proceedings were excluded. Finally, RCTs published in languages other than English were excluded.

After removing duplicates, the article titles and abstracts were screened by 2 independent reviewers. These reviewers flagged any publication that may be relevant and downloaded the full text. During the full-text review, reviewer disagreement was resolved by consensus.

Data Extraction

The following research characteristics were extracted: publication year, recruitment period, number of randomized patients, number of asymptomatic patients, and median follow-up duration.

The following clinical and demographic characteristics were also extracted: gender, age, degree of stenosis at randomization, vascular risk factors, and medications before randomization. Data on the incidence of any long-term or perioperative stroke, MI, and death were extracted. If none of these results were provided when asymptomatic patients were recorded in the text, we contacted the corresponding author to solicit these data as possible. The perioperative period was defined as 30 days after the operation. The long-term period was defined as the entire period after surgery (including the perioperative period).

Quality Assessment

We used the Cochrane Collaboration risk of bias assessment tool to assess research quality. Specifically, each RCT was evaluated for random sequence generation, allocation concealment, result evaluation blinding, result data completeness, selective result reporting, and other sources of bias. The role of funding sources was also considered ([Figure S1–S2](#)). For each bias domain, 2 independent reviewers assigned scores for unclear, low, or high risk of bias. Disagreements were resolved by consensus or by a third reviewer when necessary. Regardless of the quality of the evaluation, RCTs were included for systematic review and meta-analysis. When asymptomatic carotid artery stenosis represented a subgroup of a given RCT, the quality of the entire RCT was assessed.

Statistical Analysis

To compare the incidence of perioperative outcome events between CAS and CEA, we used the Mantel-Haenszel method fixed-effects model to estimate odds risk (OR) and 95% CI. Heterogeneity was assessed with the I^2 statistic. For evaluating long-term outcome, we used the graphical data extraction software Engauge Digitizer 4.1 to continuously extract the survival rates of CEA or CAS populations at different time points from the Kaplan-Meier curves of the original literature and exported them into ".csv" format files according to previous literature of Guyot et al.⁷ The obtained data were used to calculate log hazard ratio, standard error of log hazard ratio, observed-expected events research (O-E), and variance

value according to the method reported by Tierney et al.⁸ $\text{Exp}[(\text{O-E})/\text{Var}]$ fixed effect modeling was used to calculate the Peto OR. In addition, a funnel chart was constructed to assess the risk of publication bias for perioperative composite end point events (Figure S3). All statistical analysis was performed using Review Manager 5.4.1.

RESULTS

Our electronic search revealed 3821 potentially relevant publications. After removing duplicates, the titles and abstracts of 3725 publications were screened. A total

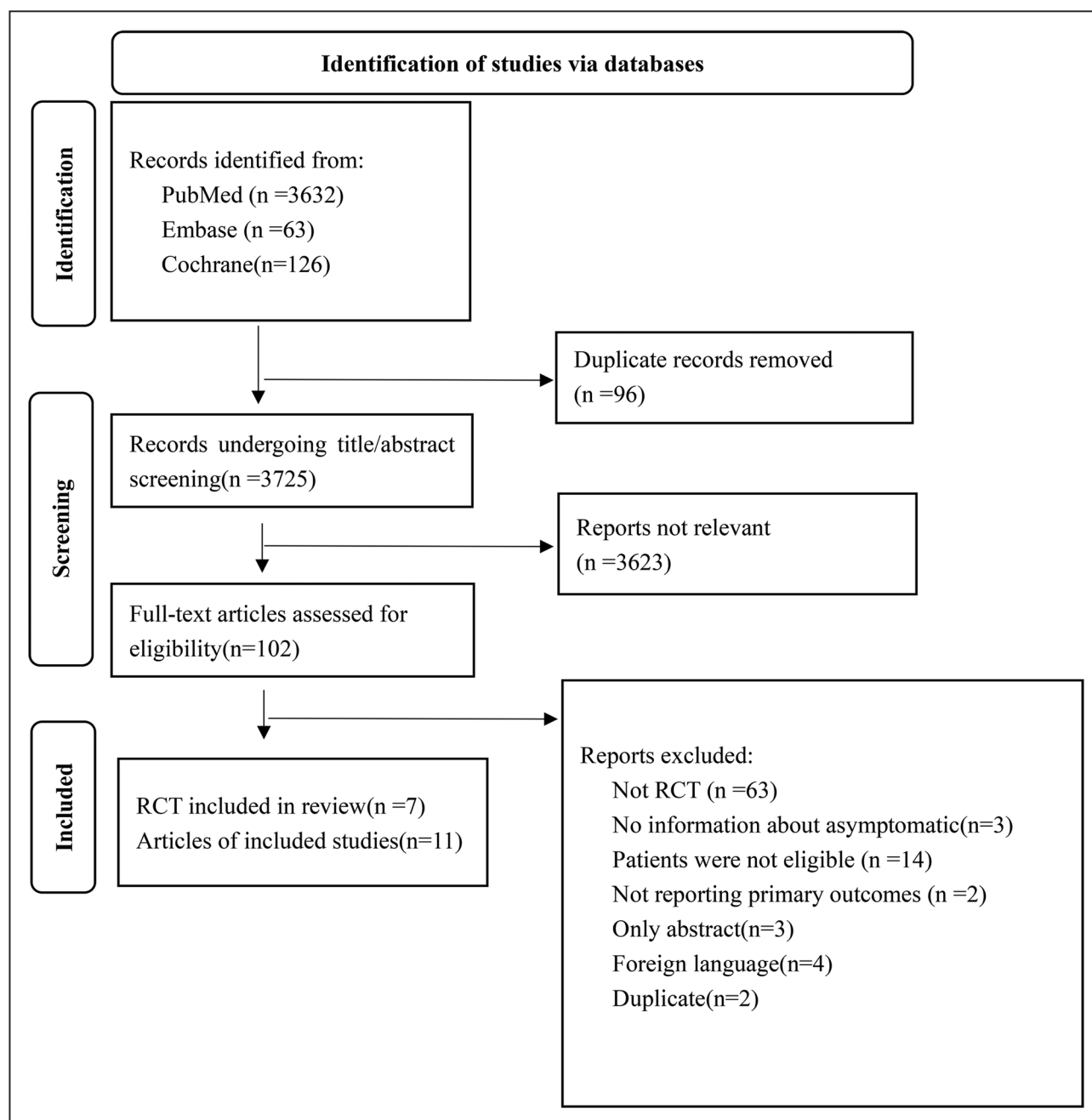


Figure 1. PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram of included randomized controlled trials (RCTs) comparing carotid artery stenting to carotid artery endarterectomy in asymptomatic patients.

of 102 full texts were downloaded. Sixteen articles from 7 RCTs^{5,9-23} reported relevant results for asymptomatic carotid artery stenosis patients (Figure 1).

Study Characteristics

The 7 included RCTs randomized a total of 8752 patients, among whom 7118 were asymptomatic (3920 randomized to CAS and 3198 randomized to CEA; Table 1).

Patient Characteristics

The included RCTs reported relatively similar populations. The average age was between 66.6 and 70.0 and the proportion of women was between 30.0% and 43.1%. In 6 trials that reported stenosis degree $\geq 95.8\%$ of patients harbored stenoses $\geq 70\%$. Regarding vascular risk factors, 36.0% to 97.6% patients had hypertension; 11.9% to 49.0% patients had diabetes; 35.0% to 81.4% patients had coronary heart disease. The proportion of smokers was between 22.0% and 93.0%. In each trial, patients were treated with aspirin and/or clopidogrel before surgery.^{5,10,11,18,20,23,24}

Quality Assessment

The overall risk of bias among included RCTs was low. Most studies received a low risk of bias for blinding of participants, blinding of outcome assessment, incomplete outcomes data, and selective reporting. Notably, the SAPHIRE trial (Stenting and Angioplasty With

Protection in Patients at High Risk for Endarterectomy) was considered to have a high risk of bias for random sequence generation as some patients were assigned at the discretion of clinicians rather than randomly. If a surgeon concluded that CEA could not be safely performed but an interventionalist judged that CAS was feasible, the patient was assigned to CAS. Likewise, if a surgeon deemed the patient suitable for surgery but the interventionalist did not feel that stenting was feasible, the patient was entered into a surgical registry.²⁰ All studies received an unclear risk of bias on random sequence generation except ACST-2 and CREST (The Carotid Revascularization Endarterectomy Versus Stenting Trial), which were judged to have a low risk of bias for this domain. We were unable to judge the risk of bias for some studies in the domains of allocation concealment^{10,12,20,23} and selective reporting.¹⁰

Periprocedural Outcomes

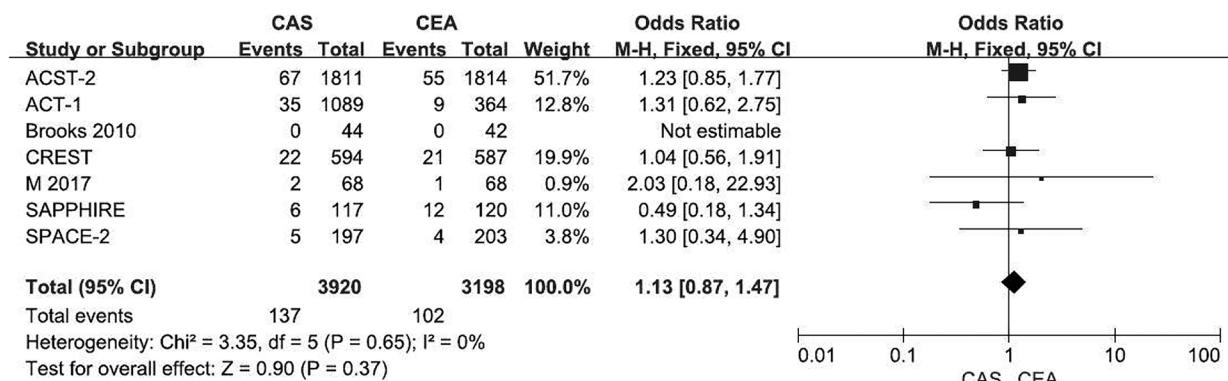
An investigational team reviewed the patient 1 month afterward and described any periprocedural events in all included RCTs.^{5,10,11,18,20,23,24} Compared with CEA group, the CAS group had no difference in perioperative composite end point events, including stroke, death, and MI (7 trials^{5,10-12,18,20,21} OR, 1.13 [95% CI, 0.87–1.47]; $P=0.37$, $I^2=0\%$). Compared with CEA, CAS had a higher risk of any stroke during the perioperative period (5 trials^{4,8,9,15,20} OR, 1.62 [95% CI, 1.16–2.24]; $P=0.004$, $I^2=0\%$) and an increased risk of nondisabling stroke (4 trials^{5,10,11,18} OR, 1.81 [95% CI, 1.23–2.65]; $P=0.003$, $I^2=0\%$). But

Table 1. Baseline Characteristics of Selected Articles

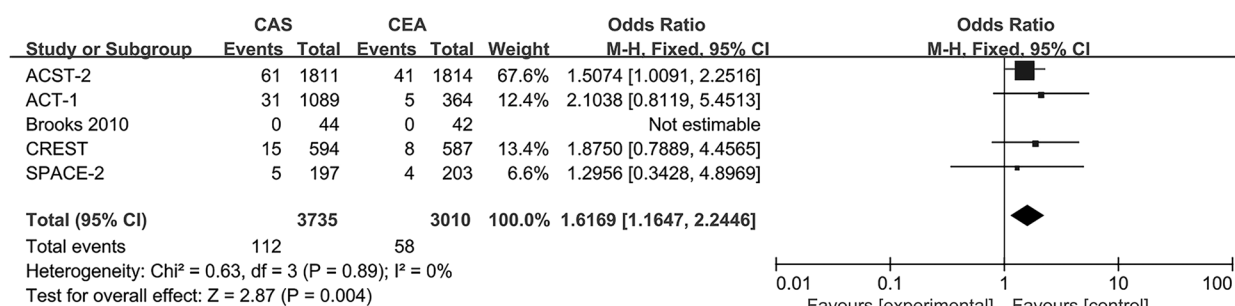
Trial	Publication year	Recruitment year	Population	Follow-up	Total	Asymptomatic		Randomized patients	Age, mean (sd)	Male (%)	Ipsilateral stenosis $\geq 70\%$
ACST-2 ⁵	2021	2008–2020	Asymptomatic with $\geq 60\%$ stenosis	5 y	3625	3625	CAS	1811	NR	1272 (70)	1751 (97)
							CEA	1814	NR	1273 (70)	1756 (97)
ACT-1 ¹¹	2016	2005–2013	Asymptomatic with $>70\%$ stenosis	5 y	1453	1453	CAS	1089	67.7 (7.0)	666 (61.2)	1089 (100)
							CEA	364	67.9 (6.9)	207 (56.9)	364 (100)
Brooks ^{14,21,23}	2001	1998–2002	Symptomatic with $>70\%$ stenosis or asymptomatic with $>80\%$ stenosis	10 y	189	85	CAS	44	66.6	NR	44 (100)
							CEA	42	69.9	NR	42 (100)
CREST ^{13,15,16,18}	2016	2005–2008	Asymptomatic with $>70\%$ stenosis	10 y	2502	1181	CAS	594	69.0 (8.0)	379 (63.8)	551 (92.8)
							CEA	587	69.6 (8.1)	396 (67.5)	539 (91.8)
M2017 ¹⁰	2017	NR	Asymptomatic with $>70\%$ stenosis	2 y	136	136	CAS	68	69 (7)	45 (66)	68 (100)
							CEA	68	68 (8)	48 (71)	68 (100)
SAPHIRE ^{19,20}	2004	2000–2002	Symptomatic with $>50\%$ stenosis or asymptomatic with $>80\%$ stenosis	3 y	334	237	CAS	117	NR	NR	117 (100)
							CEA	120	NR	NR	120 (100)
SPACE-2 ^{9,12,24}	2019	2009–2014	Asymptomatic with $>50\%$ stenosis	1 y	513	400	CAS	197	69 (8)	143 (73)	NR
							CEA	203	70 (7)	151 (74)	NR

ACST-2 indicates Second Asymptomatic Carotid Surgery Trial; ACT-1, Asymptomatic Carotid Trial; CAS, carotid artery stenting; CEA, carotid endarterectomy; CREST, The Carotid Revascularization Endarterectomy Versus Stenting Trial; SAPHIRE, Stenting and Angioplasty With Protection in Patients at High Risk for Endarterectomy; and SPACE-2, Stent-Protected Angioplasty in Asymptomatic Carotid Artery Stenosis vs Endarterectomy.

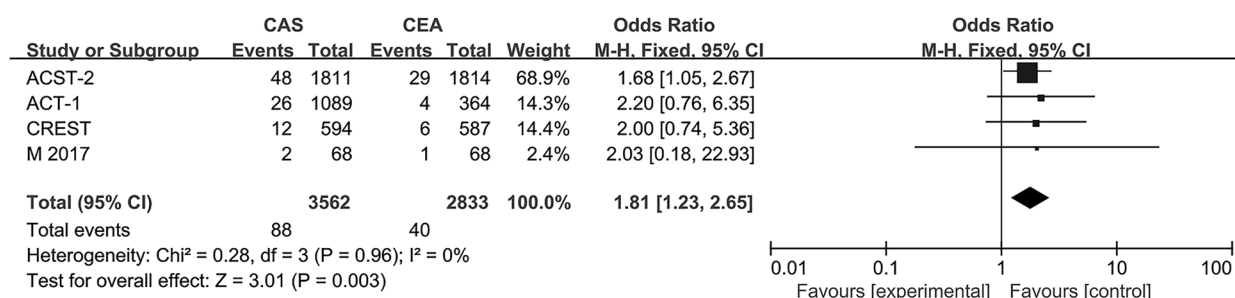
30 days outcome of composite endpoint events including stroke, death and MI



30 days outcome of any stroke



30 days outcome of non-disabling strokes



30 days outcome of disabling strokes and death

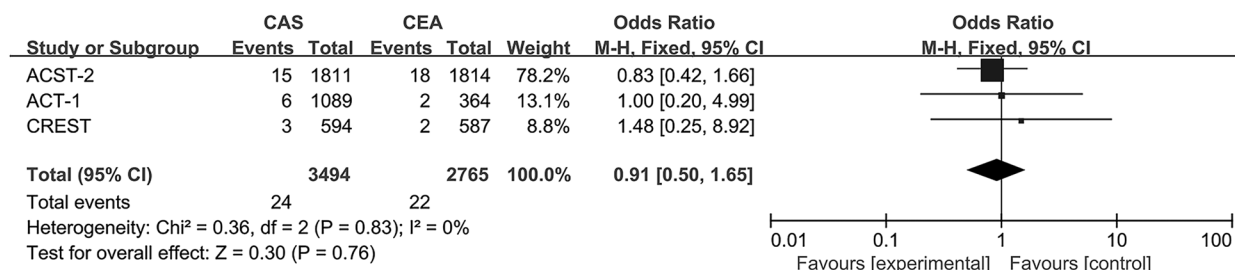


Figure 2. Forest plots of periprocedural (30-d) outcomes in randomized controlled trials comparing carotid artery stenting (CAS) vs carotid artery endarterectomy (CEA) in patients with asymptomatic carotid stenosis.

ACST-2 indicates Second Asymptomatic Carotid Surgery Trial; ACT-1, Asymptomatic Carotid Trial; CREST, The Carotid Revascularization Endarterectomy Versus Stenting Trial; M-H, Mantel-Haenszel; MI, myocardial infarction; SAPPHIRE, Stenting and Angioplasty With Protection in Patients at High Risk for Endarterectomy; and SPACE-2, Stent-Protected Angioplasty in Asymptomatic Carotid Artery Stenosis vs Endarterectomy.

there was no significant difference in disabling stroke and death between groups (3 trials^{5,11,18} OR, 0.91 [95% CI, 0.50–1.65]; $P=0.76$, $I^2=0\%$; Figure 2).

Long-Term Outcomes

In ACST-2, annual follow-up for monitoring any strokes were by the ACST office writing to the patient and the events were reported by the patient on a follow-up form sent by mail. After the 1-month postprocedural form, no further follow-up by the doctor is required.⁵ Except ACST-2, follow-up for monitoring any strokes were reported by an investigational team to perform neurological assessment in other RCTs.^{10,11,18,20,23,24} Kaplan-Meier curve data were extracted from 4 RCTs involving 5564 asymptomatic carotid artery stenosis patients.^{5,9,11,14} Compared with CEA, there was no risk difference in the composite outcome among any perioperative stroke, death, or MI and long-term stroke, death, or MI (Peto OR, 1.18 [95% CI, 0.94–1.48]; $P=0.14$, $I^2=0\%$; Figure 3).

GRADE Assessment and Summary of Findings Tables

We used the GRADEpro Guideline Development Tool to create summary of findings tables (Table 2) for the comparison of stenting versus endarterectomy for asymptomatic carotid artery stenosis. Table 2 present the results and the certainty of the evidence of the main outcomes, using the GRADE system (The Grading of Recommendations Assessment, Development and Evaluation), which classifies the certainty of the evidence as high, moderate, low, and very low.

DISCUSSION

Our study compared the safety and efficacy between CAS and CEA in treating asymptomatic carotid artery stenosis. The composite end point at 30 days after treatment was not significant different between CAS and CEA. Although CAS showed a higher incidence of any or nondisabling stroke during the perioperative period,

there was no significant difference in disabling stroke and death between the 2 groups. For long-term outcomes, no difference in the composite outcome of any stroke, death, and MI existed between CEA and CAS.

There are several trials showing CEA and CAS have comparable safety when treating asymptomatic carotid artery stenosis. For example, for the composite end point 30 days after the procedure, stenting was noninferior to endarterectomy (event rates, 3.8% versus 3.4%, $P=0.60$) in ACT-1.¹¹ CREST also showed no significant difference between CAS and CEA (7.2% versus 6.8%; hazard ratio, 1.11 [95% CI, 0.81–1.51]; $P=0.51$).¹⁶ A pooled analysis²⁵ found that any periprocedural stroke (RR, 1.84 [95% CI, 0.99–3.40]), periprocedural nondisabling stroke (RR, 1.95 [95% CI, 0.98–3.89]), and any periprocedural stroke or death (RR, 1.72 [95% CI, 0.95–3.11]) trended toward an increased risk after CAS. Compared with CEA, CAS showed a higher 30-day death/stroke rate in a previous meta-analysis involving 3467 asymptomatic carotid artery stenosis.²⁶ Also, the latest Cochrane systematic review showed a trend toward higher incidence of death or any stroke during the periprocedural period with CAS.⁴ Our results were similar to the aforementioned studies, and CEA may have superiority over CAS in reducing any or nondisabling stroke in asymptomatic carotid artery stenosis, but no difference in composite end point during perioperative period. This may be the attributable to the inclusion of MI in the composite end point, which created the potential for the higher risk of minor stroke associated with CAS to be offset by the higher risk of MI associated with CEA. Statistical differences from previous studies may be attributable to the inclusion of ACST-2, the largest RCT in asymptomatic carotid artery stenosis, which weighted 72.8% in meta-analysis.

There are several trials reporting that CEA and CAS have similar long-term outcomes. ACT-1 did not find a significant difference in long-term outcomes between the study groups.¹¹ Similarly, CREST showed no significant difference over 10-years of follow-up.¹³ The total event numbers were 8.5% CAS versus 7.05% CEA during follow-up in ACST-2 ($P=0.09$).⁵ A meta-analysis recruiting 3 trials involving 2,871 asymptomatic carotid

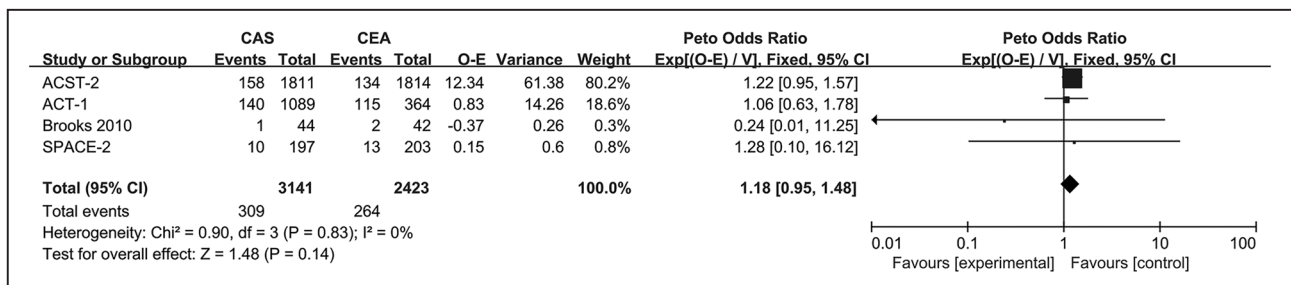


Figure 3. Forest plots of long time outcomes of composite outcome among any perioperative stroke, death, or myocardial infarction (MI) and long-term stroke, death, or MI in randomized controlled trials comparing carotid artery stenting (CAS) vs carotid artery endarterectomy (CEA) in patients with asymptomatic carotid stenosis.

ACST-2 indicates Second Asymptomatic Carotid Surgery Trial; ACT-1, Asymptomatic Carotid Trial; and SPACE-2, Stent-Protected Angioplasty in Asymptomatic Carotid Artery Stenosis vs Endarterectomy.

Table 2. Summary of Findings for the Main Comparison

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Certainty of the evidence (GRADE)
	Risk with CEA	Risk with CAS			
Long-term outcomes	109 per 1000	129 per 1000 (104–161)	Peto OR, 1.18 (0.95–1.48)	5564 (4 RCTs)	⊕⊕⊕⊕ High
30 days composite outcomes	32 per 1000	36 per 1000 (28–46)	OR, 1.13 (0.87–1.48)	7118 (7 RCTs)	⊕⊕⊕⊕ High
30 days outcome of any stroke	19 per 1000	31 per 1000 (22–42)	OR, 1.62 (1.16–2.24)	6745 (5 RCTs)	⊕⊕⊕⊕ High
30 days outcome of nondisabling stroke	14 per 1000	25 per 1000 (17–37)	OR, 1.81 (1.23–2.65)	6395 (4 RCTs)	⊕⊕⊕⊕ High
30 days outcome of disabling stroke	8 per 1000	7 per 1000 (4–13)	OR, 0.91 (0.50–1.65)	6259 (3 RCTs)	⊕⊕⊕⊕ High

GRADE Working Group grades of evidence: High certainty: we are very confident that the true effect lies close to that of the estimate of the effect. Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect. Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect. The table is produced by the GRADEpro online tool (<https://gdt.gradepro.org/app/>). CAS indicates carotid artery stenting; CEA, carotid endarterectomy; OR, odds ratio; and RCT, randomized controlled trial.

*The risk in the intervention group (and its 95% CI) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

artery stenosis patients could not establish the clinical superiority of either treatment regarding long-term outcomes (RR, 0.92 [95% CI, 0.70–1.21]).⁵ The latest Cochrane systematic review showed the composite end point until last follow-up did not differ between groups (2 trials involving 397 asymptomatic carotid artery stenosis patients, RR, 0.84 [95% CI, 0.48–1.47]).⁴ Compared with these studies, ours has the advantage of including a greater number of trials and patients, restricted to large RCTs. For long-term outcomes, our results showed no difference in the composite outcome of any stroke, death, and MI existed between CEA and CAS.

However, for asymptomatic patients with stenosis $\geq 50\%$, the annual stroke rate is 0.5% to 1.0%,^{18,21} which was much lower than that of symptomatic patients. Therefore, is revascularization necessary for asymptomatic carotid artery stenosis? Based on a meta-analysis comparing CEA with BMT in asymptomatic carotid artery stenosis, CEA had worse short-term outcomes but more favorable long-term outcomes in this population.²⁵ However, the period of patient recruitment was mostly before 2000.^{22,27,28} In the past 20 years, BMT has also undergone considerable innovation. Meanwhile, CEA will likely always have some degree of perioperative complications, but the direct control over embolization is desirable. BMT will be often the safest short-term option, but the risk of long-term embolism will likely invariably accumulate. When it comes to asymptomatic carotid artery stenosis, comparison between BMT and revascularization in the short-term is likely meaningless and these 2 methods should be compared with annual stroke rates and long-term prognosis. We now understand the need for new trials to compare BMT with CAS and CEA in contemporary practice. Two such trials (SPACE-II [Stent-Protected Angioplasty in Asymptomatic Carotid Artery Stenosis vs Endarterectomy]²⁹ and CREST-II²⁴) are currently underway and will provide important information on the comparative efficacy and safety of BMT alone and CAS plus BMT or CEA plus BMT.

There are some limitations of the present work. Most of the trials included in this review did not provide a detailed description of the BMT regimen, so it is not possible to conduct a subgroup analysis comparing the changes in CAS or CEA benefits brought about by improvements in BMT regimens over the past decades. Studies published in non-English language were excluded. We call for more native speakers of non-English language to conduct similar searches in future studies. Also, most of the trials were restricted to patients in whom intervention was considered necessary and did not include specific subgroups of patients who were at high risk for complications from endarterectomy or stenting. Thus, these participants are not representative of the wide spectrum asymptomatic patients. In the future, risk-stratification techniques should be used to select more appropriate asymptomatic carotid artery stenosis patients to receive revascularization. We could not analyze all the subtypes of end points done for 30 days for long term as well limited by insufficient data from the original study. Thus, we called for individual participant data meta-analysis. Individual participant data meta-analysis is a specific type of systematic review, in which rather than extracting summary (aggregate) data from study publications or from investigators, the original research data are sought directly from the researchers responsible for each study.

CONCLUSIONS

When treating asymptomatic carotid artery stenosis, CAS has comparable perioperative and long-term composite outcomes compared with CEA. However, CAS may have a higher risk of any stroke and nondisabling stroke during the perioperative period.

ARTICLE INFORMATION

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Disclosures

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Supplemental Material

PRISMA Checklist

Tables S1

Figure S1–S3

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